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1- Introduction:

Explanation for the Entity-Relationship Diagram (ERD) and database design used in the Fitness Club Management System are provided in this document. The purpose of the ERD is to model how members, trainers, classes, training sessions, bookings, and health metrics are stored and related inside PostgreSQL.

2- Entities Overview:

- Member: Stores essential personal details for gym members.
- Trainer: Contains information about fitness trainers and instructors.
- Room: represents workout rooms used for classes or sessions.
- FitnessGoal: Records goals created by members.
- HealthMetric: Logs measurements such as weight, height, and heart rate.
- PersonalTrainingSession: One-on-one sessions between members and trainers.
- GroupClass: Group fitness classes led by an instructor.
- ClassEnrollment: The linking table for members enrolled in group classes.
- TrainerAvailability: Stores the availability of trainers.

- Equipment: Gym equipment associated with rooms.
- Billing: Payment history and charges for member accounts.

3- Relationships:

- A Member can have many FitnessGoals.
- A Member can log many HealthMetrics over time.
- A Member can book multiple PersonalTrainingSessions.
- A Trainer can handle many PersonalTrainingSessions.
- A Room can host many sessions and group classes.
- A Trainer teaches many GroupClasses.
- Members join GroupClasses through ClassEnrollment.
- Trainers have multiple availability slots in TrainerAvailability.
- Rooms contain multiple Equipment items.
- Members may have multiple Billing entries.

4- Normalization:

- 1NF: Every attribute is atomic with no repeating groups.
- 2NF: All non-key attributes fully depend on the table's primary key.
- 3NF: No transitive dependencies; related information is placed in separate tables using foreign keys.

5- Relational Schema Summary (Examples):

- Member(member_id PK, first_name, last_name, email, phone, ...)
- Trainer(trainer_id PK, first_name, last_name, email, ...)
- PersonalTrainingSession(session_id PK, member_id FK, trainer_id FK, ...)
- GroupClass(class_id PK, trainer_id FK, room_id FK, ...)
- ClassEnrollment(enrollment_id PK, member_id FK, class_id FK)

6- Advanced SQL Features That Improve Reliability And Efficiency:

- Triggers to prevent overlapping trainer sessions.
- Indexes to improve performance when searching or filtering.
- A view that simplifies dashboard queries.

7- Conclusion:

The ERD and relational schema provide a clean, normalized, and scalable structure for the Fitness Club Management System. The database supports all main features and forms a strong foundation for the application.